

Elsa Elisa Yohana Sianturi

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Informatics Engineering graduate focused on AI Engineering, Natural Language Processing, Computer Vision, and Multimodal Learning. Experienced in collecting, preparing, and annotating custom datasets with LLM assistance, as well as building RAG-based document question-answering systems and multimodal classifiers. Skilled in Python, PyTorch, Hugging Face, Scikit-learn, OpenCV, and LLM integration, with strong attention to data quality and model evaluation

EDUCATION

Bachelor's Degree in Informatics Engineering | Institut Teknologi Sumatera

Aug 2022 - May 2026

Lampung, Indonesia

- GPA: 3.64/4.00
- Focus: Deep Learning, Machine Learning, NLP, and Computer Vision
- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Multimodal Machine Learning, Data Visualization and Information

EXPERIENCE

AI Engineer Intern — PT Pertamina Hulu Rokan

Jun 2025 - Jul 2025

Pekanbaru, Indonesia

- Developed AIDocs, a RAG-based question-answering prototype that accelerates information retrieval from PDF land agreements, eliminating the need for manual document checks
- Engineered the end-to-end data pipeline, covering document preprocessing, hybrid retrieval, answer generation, and source citation utilizing LangChain, FAISS, and BM25
- Evaluated and optimized system performance against original document content, achieving 73.33% answer accuracy and successfully delivering an AI-driven solution for document digitalization

Teaching Assistant — UPA TIK ITERA

Aug 2023 - Present

Lampung, Indonesia

- **Operating Systems**
Facilitated hands-on laboratory sessions focusing on process management, memory allocation, and operating system fundamentals
- **Introduction to Computing & Software I (PKS I)**
Guided students in procedural programming, computational thinking, and fundamental programming logic
Credential ID: B/5359/IT9.E2/TA.00.00/2024
- **Interaction Design**
Provided technical mentoring for student UI/UX projects, user-centered design principles, and project-based assignments
Credential ID: 33133/IT9.3.3/TA.00.00/2025

PROJECTS

Multimodal Deep Learning for Self-Harm Meme Detection | [GitHub]

Aug 2025 - Apr 2026

Tech: CLIP, ELECTRA, Gradio

- Built and curated a custom dataset of 2,024 memes by collecting and cleaning image-text samples, then performing LLM-assisted annotation for self-harm and non-self-harm labels based on predefined criteria
- Developed a multimodal binary classification system for a final thesis project to identify self-harm versus non-self-harm memes using combined visual and textual representations
- Trained and evaluated the model on a stratified dataset of 2,024 memes, achieving an outstanding 96.1% F1-score and successfully outperforming both single-modality baselines (CLIP at 91.5% and ELECTRA at 92.0%)

Multimodal City Branding Analysis | [GitHub]

May 2026 - Present

Tech: Image-Text Representation Learning, Social Media Analysis

- Developed a multimodal deep learning model to analyze South Lampung tourism image and caption data from Instagram
- Achieved approximately 89.2% validation accuracy at the selected best epoch based on validation loss

- Collaborated with researchers and students from Urban and Regional Planning and Informatics Engineering

Student Face Recognition with ArcFace and SVM | [GitHub]

Jun 2026

Tech: InsightFace ArcFace, Scikit-learn, OpenCV, SVM, Jupyter Notebook

- Developed an end-to-end facial identification system to automate the recognition of 70 student identities from uploaded facial images
- Engineered a pipeline using offline image augmentation, 512-dimensional InsightFace ArcFace Buffalo_L embeddings, and an RBF-kernel SVM classifier with leakage-resistant Stratified Group K-Fold evaluation
- Processed 1,946 facial samples and achieved 82.0% accuracy, 85.4% weighted precision, and 82.4% weighted F1-score

LEADERSHIP & ORGANIZATIONS

Head of Academic & Scholarship Division — HMIF ITERA

Jan 2025 – Dec 2025

Lampung, Indonesia

- Led 4 academic and scholarship-related programs, served as a speaker in a scholarship-sharing session, and managed mentoring initiatives that contributed to 2 students receiving the Pertamina Sobat Bumi Scholarship

Secretary of Minister of Information Technology — KM ITERA

Apr 2024 – Dec 2024

Lampung, Indonesia

- Managed meeting notes, project documentation, and GitHub-based timeline tracking to coordinate IT-related student government web projects and report progress to the Minister of Information Technology

Head of Entrepreneurship Department — Sobat Bumi Lampung

Oct 2025 – Present

Lampung, Indonesia

- Coordinated vendor communication and production planning for Sobat Bumi Lampung merchandise, including jackets, vests, and community attributes, while supporting entrepreneurship initiatives within the scholarship community

SKILLS

- **Programming:** Python, SQL
- **Data Preparation & Quality:** Data Collection, Data Cleaning, LLM-assisted Annotation, Dataset Curation
- **Machine Learning & Deep Learning:** PyTorch, Scikit-learn, Hugging Face, OpenCV, Pandas, NumPy, Matplotlib
- **NLP & LLM:** LangChain, FAISS, BM25, ChromaDB, OpenAI API, RAG, Transformer-based Models
- **Computer Vision:** CLIP, EfficientNet-B8, ResNet-50, Image Classification, Multimodal Learning
- **Model Development:** Fine-tuning, Hyperparameter Tuning, Ablation Study, Error Analysis, Model Evaluation
- **Deployment & Tools:** Gradio, RunPod, Hugging Face Spaces, FastAPI, Git/GitHub, Linux, W&B, Figma, Tableau

CERTIFICATIONS

- **AWS re/Start Cloud Computing Program** - Orbit Future Academy & Amazon Web Services (Mar 2026 - Present)
- **Belajar Pemrograman Prosedural dengan Python** - Dicoding (2024) | Credential ID: 2VX3O4W23ZYQ
- **Memulai Pemrograman dengan Python** - Dicoding (2024) | Credential ID: L4PQQ8O2PO01
- **Belajar Dasar Structured Query Language (SQL)** - Dicoding (2024) | Credential ID: NVP77M3WVPR0
- **AI Praktis untuk Productivity** - Dicoding (2024) | Credential ID: 1OP82QOYLPQK

AWARDS & HONORS

- **Pertamina Sobat Bumi Scholarship Recipient** - Pertamina Foundation (2024)
- **2nd Winner of UI/UX Competition Point Project 2.0** - HMIF ITERA (2024) | Credential ID: 313/525/SRT/HMIF/XII/2024

LANGUAGES & PUBLICATIONS

- **Languages:** Indonesian (Native), English (Professional Working Proficiency)
- **Publications:** Sianturi, E. E. Y., et al. "Transformasi Digital Layanan Posyandu." PADIMAS, 2025 | [Full Paper]